

MSc in Data Science

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Capstone Project: Identifying Identical and Variant Products in E-commerce and Visual Feature Extraction for Fashion Items

A Case Study with Jinius Marketplace

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Abstract

This study presents a comprehensive framework for improving product catalog organization in online marketplaces through automated detection of identical and variant listings, as well as visual feature extraction for fashion items. Conducted in collaboration with the Jinius Marketplace in Cyprus, the project addressed **three primary objectives**: identifying and merging duplicate product listings, grouping product variants, and extracting descriptive visual attributes from dress images.

Multiple methodologies were explored for product and variant detection, ranging from metadata-based grouping and embedding-driven clustering to **multi-modal fusion techniques** integrating textual and visual representations. A hybrid approach combining Sentence-BERT contextual embeddings with ResNet50 and CLIP image embeddings, clustered using HDBSCAN, achieved near-perfect results, with cluster purity exceeding 99% and micro-averaged precision reaching 97%. These findings demonstrate the effectiveness of multimodal embedding strategies in handling noisy and inconsistent marketplace data.

For visual feature extraction, the CLIP vision-language model was employed to perform zero-shot attribute recognition on fashion imagery. Prompt engineering and YOLO-based garment localization significantly improved the interpretability and consistency of results, enabling the model to identify attributes such as sleeve style, neckline, and pattern without supervised training.

The outcomes of this project not only enhanced product grouping and attribute extraction accuracy but also provided a scalable and production-ready pipeline that has been integrated into the Jinius Marketplace platform. This integration improves search efficiency, catalog coherence, and user experience, establishing a foundation for future development of intelligent product organization systems in e-commerce environments.

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Chapter 1

Introduction

Online marketplaces have transformed global commerce by enabling consumers to access a wide variety of products from multiple merchants. Despite this convenience, such platforms face persistent challenges in terms of product consistency and catalog organization. Duplicate listings of the same product, often uploaded by different sellers, and variant listings of the same model differing only in attributes such as color or size, are common issues [1]. These inconsistencies reduce the clarity of product catalogs, complicate product discovery, and can negatively affect both user trust and platform efficiency. As noted by Gu et al. [2], fashion data is inherently multimodal, noisy, and often inconsistently annotated across platforms, making reliable product understanding and organization a persistent challenge for AI-driven systems.

The present study, undertaken in collaboration with the Jinius online marketplace in Cyprus, addressed three interconnected tasks: (1) the identification and grouping of identical products, (2) the detection and clustering of product variants, and (3) the extraction of product features from images to improve product representation. A range of methodologies were developed and evaluated, combining heuristic-based grouping with modern machine learning and computer vision techniques.

For **Tasks 1 and 2**, which focused on duplicate and variant product grouping, four approaches were explored. The **initial method** applied **group-by operations** to existing product attributes, such as brand and color. Although straightforward, this strategy was limited by incomplete and inconsistent data fields. More advanced methods incorporated **deep learning embeddings**, including ResNet50 image embeddings [3] and TF-IDF representations of product attributes, which were **combined using weighted averages** and clustered using density-based algorithms such as DBSCAN and HDBSCAN. Additional experiments included **concatenation** and **element-wise multiplication of image and text embeddings**, followed by clustering with **DBSCAN, HDBSCAN, K-means**, and **agglomerative clustering** [4]. The most effective approach employed **contextual embeddings** from **Sentence-BERT** [5], integrated with image embeddings, and clustered using HDBSCAN. This final method achieved near-perfect performance when evaluated against a manually curated ground truth of product variants.

Task 3 concentrated on feature extraction within a dress dataset, where the objective was to derive meaningful attributes directly from product images. The **CLIP (Contrastive Language–Image Pretraining) model** [6] was applied to **align images and natural language prompts** in a shared embedding space, enabling **zero-shot** attribute classification. Experiments compared short, generic prompts with longer descriptive ones, showing that shorter prompts often produced slightly better results. To further improve performance, **YOLO (You Only Look Once) algorithm** [7] was introduced as an **object detection step** to crop images around the relevant product region before passing them to CLIP, thereby reducing background noise and improving recognition accuracy. Research in fashion AI shows that fashion images contain rich and complex multimodal signals, and that visual–textual inconsistency is a core challenge in fashion applications [2].

Taken together, these methodologies form a scalable framework for addressing duplicate and variant product listings while simultaneously enriching product metadata through computer vision. The integration of clustering algorithms, multimodal embeddings, and automated feature extraction contributes both to enhanced user experience and to greater operational efficiency within online marketplaces.

Chapter 2

Background

Online marketplaces such as *Jinius*, a leading e-commerce platform in Cyprus, provide access to a wide range of products from multiple merchants. While this variety benefits consumers, it also introduces significant challenges in data consistency and catalog management. As noted by Akritidis et al. [1], the same product often appears multiple times in slightly different forms, either as duplicate listings, identical items uploaded by different sellers or as variants of the same model offered in different colors, sizes, or by multiple merchants.

The presence of duplicates and fragmented variants negatively affects both **user experience and platform efficiency**. For consumers, redundant listings create a cluttered interface, requiring additional effort to identify the desired product. This not only slows down product discovery but also reduces trust in the platform if price or attribute information appears inconsistent. Rubio et al. [4] and McAuley et al. [8] emphasize that, from the platform’s perspective, duplicate entries dilute engagement metrics, complicate inventory management, and weaken the performance of search and recommendation systems.

To address these challenges, several studies, including Graczyk et al. [9], propose the use of product grouping and deduplication systems. Such systems aim to identify identical products, cluster their variants under a unified model, and display them as a single consolidated entry with options for size, color, or seller. This strategy improves catalog clarity, simplifies the shopping process, and enhances the operational efficiency of tasks such as price comparison and inventory tracking.

A key component of these systems is feature extraction through computer vision. This aligns with the analysis of Gu et al. [2], who highlight that fashion images contain complex semantic cues—such as garment shape, material, neckline, and fine-grained attributes—while also being affected by pose variation, occlusion, and background noise, making visual feature extraction essential for fashion analysis. Unlike textual metadata, which is often incomplete, inconsistent, or multilingual, product images contain highly discriminative information about an item’s visual identity. Advances in convolutional neural networks (CNNs) have enabled the extraction of robust visual features such as color, texture, and shape, as demonstrated by He et al. [3] and Krizhevsky et al. [10]. These image embed-

dings can be used to cluster visually similar products and identify variants more effectively. Recent research by Varnukhov et al. [11] and Song et al. [12] further shows that combining visual and textual embeddings through multimodal learning substantially improves product-matching accuracy.

Addressing the issue of duplicate and variant product listings is therefore crucial for the long-term scalability of online marketplaces. By integrating clustering algorithms, multimodal feature representations, and computer vision techniques, platforms can enhance catalog coherence, improve user experience, and achieve greater operational efficiency in large-scale e-commerce environments.

Beyond the challenge of identifying duplicates and variants, marketplaces also benefit from extracting descriptive visual attributes directly from product images. Attributes such as sleeve length, neckline, pattern, or dress occasion add interpretability and improve catalog organization, yet are often missing or inconsistently filled in textual metadata. Recent advances in vision-language models, such as CLIP [6], allow zero-shot extraction of such attributes by aligning text and image embeddings without the need for labeled data. Incorporating this capability complements the clustering work of Tasks 1 and 2 by enriching grouped products with meaningful, image-derived attributes, ultimately enhancing both search relevance and the overall user experience.

Chapter 3

Collaboration with Jinius Marketplace

Jinius Marketplace is a digital commerce platform designed to support both consumers and merchants by providing a unified space for product discovery, online retail, and service facilitation. As part of the Bank of Cyprus Group, the platform aims to strengthen the local digital economy by enabling small and medium-sized businesses to reach broader audiences through streamlined digital tools. Jinius offers a range of services including product catalog management, merchant onboarding, logistics support, and customer engagement features. Its infrastructure accommodates a diverse catalog of items from multiple sellers, making consistency, reliability, and accuracy of product information critical to creating a positive user experience.

The collaboration for this project was established to address the company's most recent needs. At the outset, the company outlined a clear set of priorities, emphasising the importance of identifying duplicate product listings, grouping product variants under unified entries, and extracting meaningful visual features from product images to improve classification and similarity analysis. These requirements shaped the scope of the research and informed the selection of appropriate machine learning and computer vision methodologies. Once the objectives were defined, Jinius Marketplace provided access to the necessary datasets, consisting of product images, metadata, and textual descriptions. This enabled the technical development to begin and ensured that the methods were designed with real-world marketplace data in mind. Collaboration was maintained through weekly progress meetings, during which initial findings were presented, challenges were discussed, and feedback was offered by the company's team. These interactions were crucial in refining both the problem definition and the methodological approach.

Beyond routine feedback, Jinius contributed valuable domain expertise throughout the project. The team shared insights into their internal systems and merchant behaviours, which helped contextualise the technical tasks and informed the interpretation of results. Their involvement ensured that the developed solutions were not only technically sound but also practically applicable within the marketplace environment.

Chapter 4

Literature Review

4.1 Product and Variant Matching

Online marketplaces face persistent challenges related to product consistency and catalog organization. These include issues such as duplicate product listings, often posted by different sellers, and variant listings, where the only differences are attributes such as size, color, or packaging. Addressing these challenges is essential for improving user experience, search efficiency, and overall platform performance. The following review examines relevant research on product matching, clustering of variants, and multimodal feature extraction, highlighting how these studies align with and inform the methodologies implemented in the present work.

Peeters et al. [13] introduced an approach to product matching that fine-tunes BERT on domain-specific product data. Their work demonstrated that intermediate pre-training of BERT substantially improves the model’s ability to identify matching products even when textual descriptions are sparse or inconsistent. The method leverages contextual embeddings to capture semantic relationships between attributes, which is directly relevant to the current study’s use of Sentence-BERT for representing textual metadata. Similarly, while Peeters et al. [13] focused primarily on enhancing textual understanding, the present work extends this concept by incorporating image embeddings derived from ResNet50 [3] and fusing them with contextual text representations to improve clustering accuracy. This multimodal approach allows for greater robustness in cases where product metadata are incomplete or partially inconsistent.

Peeters and Bizer [14] further advanced the field by proposing a supervised contrastive learning framework for product matching. Their model optimizes the embedding space by drawing semantically similar products closer together and pushing dissimilar ones apart, thereby improving differentiation among products with subtle textual variations. The concept of contrastive learning aligns with the embedding philosophy of the current study, which similarly seeks to encode semantic proximity among product variants. However, unlike Peeters and Bizer [14], who focused exclusively on textual embeddings and contrastive loss, the present study integrates visual signals through models such as ResNet50 [3] and CLIP [6], and applies clustering techniques such as DBSCAN and HDBSCAN [15] to identify

duplicates and variants without supervised labels.

Li et al. [16] examined the problem of cross-platform product matching using a dual-encoder architecture that jointly processes textual and visual data. Their framework computes similarity across modalities, enabling more effective product matching even when product metadata differ between platforms. The integration of visual and textual encoders in Li et al. [16] is conceptually similar to the multimodal approach developed in this project, where both modalities contribute to the final product embeddings. However, while their study focuses on aligning products across distinct e-commerce platforms, the current work targets the internal organization of a single marketplace, concentrating on the clustering of duplicates and variants. Building on the dual-encoder idea, this study extends the architecture by incorporating advanced unsupervised clustering algorithms such as HDBSCAN [15] to automatically discover natural groupings among marketplace items.

Shah et al. [17] proposed an extreme classification framework for large-scale product matching, employing neural networks to handle millions of product classes efficiently. Their approach demonstrates high scalability for massive catalog environments, where computational efficiency is critical. Although the marketplace context of the present study operates at a smaller scale, the emphasis on accuracy and interpretability aligns closely with Shah et al.’s [17] goals. The current work focuses on producing high-precision clustering outcomes for product variants, using computationally intensive but highly accurate methods such as contextual embedding fusion and density-based clustering. Nevertheless, the extreme classification techniques proposed by Shah et al. [17] could complement this work in future applications where marketplace data volume increases substantially.

Overall, the literature on product and variant matching has evolved through the integration of deep learning, multimodal embeddings, and advanced clustering techniques. The reviewed studies collectively emphasize the importance of semantic understanding, multimodal representation, and scalable modeling for accurate product alignment. The present study draws upon these foundations, extending them by combining contextual language models, visual representations, and clustering algorithms to enhance product deduplication and variant detection. Through this synthesis of methodologies from Peeters et al. [13], Peeters and Bizer [14], Li et al. [16], and Shah et al. [17], the work contributes to the broader development of efficient and accurate frameworks for product catalog organization in online marketplaces.

4.2 Zero-Shot Visual Attribute Extraction in Fashion Imagery

Extracting descriptive attributes (e.g., sleeve style, pattern) from fashion images is a challenging task due to variations in pose, background interference, overlapping objects, and the fine-grained nature of some attributes. Traditional su-

pervised methods rely heavily on annotated data, which is expensive to collect for every attribute and clothing style. Recent advances in vision-language models (VLMs) and zero-shot learning have introduced new methods that can infer attributes without explicit training.

CLIP is a foundational vision-language model introduced by OpenAI [6], which aligns images and textual descriptions into a shared embedding space by training on large-scale image-text pairs. Thus, it enables zero-shot classification without additional fine-tuning on specific labels.

Baron et al. [18] extended CLIP to fine-grained part-level attribute recognition through a task termed “real classification by description”, where predictions are based solely on descriptive attributes rather than object categories. Their proposed multi-resolution CLIP architecture processes multiple image crops to better detect localized features, improving performance in part-attribute recognition. While their experiments focus on general descriptive attributes, they do not explicitly include fashion-specific attributes such as sleeves or necklines, leaving a gap in applying zero-shot models to detailed garment descriptions.

Since CLIP’s predictions rely on text-image similarity, the phrasing of prompts directly affects performance. Baron et al. [18] emphasize that removing class names from prompts helps the model attend more to descriptive features. In addition to that, broader research in attribute extraction shows that structured and concise prompts (e.g., “a short sleeve dress”) outperform long, descriptive sentences, as demonstrated by Baron et al. [18], Guo et al. [19], and Patrick and Cyh [20], supporting the approach of A/B testing short versus detailed prompt sets. This reinforces the idea that prompt engineering is critical for improving zero-shot attribute classification accuracy.

As stated in [21], background noise and irrelevant image regions often reduce the reliability of attribute extraction. One practical solution is to apply object detection before feature extraction to ensure the model focuses on the garment. YOLO (You Only Look Once) is a widely used one-stage detector for this purpose [7].

Chang and Zhang [21] applied YOLOv5 to clothing style recognition, achieving high accuracy in identifying patterns such as plaid, plain, and striped. Their work demonstrates that YOLO-based cropping improves classification performance by isolating clothing regions. This concept aligns with the pipeline used in this project, where YOLO was employed to detect the person in the images and therefore crop the dress region before passing it through CLIP for attribute extraction.

From the reviewed literature, it is evident that CLIP and its extensions demonstrate strong potential for zero-shot attribute recognition, however, they have not been extensively evaluated on fashion-specific features such as sleeves, necklines, and patterns. Prompt phrasing has also been shown to play a critical role in determining model performance, where concise and attribute-focused wording generally yields more accurate and stable results than longer, descriptive phrases. Furthermore, region localization through object detection methods, such as YOLO, has proven to be an effective preprocessing step for minimizing background noise and improving the precision of attribute inference. Collectively, these findings high-

light that integrating prompt engineering and garment localization techniques can substantially enhance the robustness and interpretability of zero-shot attribute extraction models for fashion imagery.

The conducted work in Task 3 builds on these findings by integrating YOLO-based garment localization, CLIP-based zero-shot embedding comparison, and prompt A/B testing to assess how well zero-shot vision-language models can extract fashion attributes without extensive labeled datasets.

Chapter 5

Dataset Overview

5.1 Sneakers Dataset – Used for Product and Variant Matching

The dataset provided by Jinius for the first two tasks consisted of sneaker product listings extracted from the marketplace’s fashion catalog in JSON format. Each record represented a unique product listing contributed by a merchant and contained a combination of structured and semi-structured data, including metadata, product attributes, and image references.

Key attributes captured essential aspects of each listing, grouped broadly as follows:

- **Identification fields:** Unique administrative identifiers used to track products across the marketplace, such as:
 - `mirakl_product_id` – a unique product identifier across the Jinius platform;
 - `product_sku` – the merchant-specific stock-keeping unit;
 - `EAN` – the European Article Number, present for some but not all entries.
- **Descriptive attributes:** Fields defining the product’s characteristics, used later in clustering and variant detection:
 - `brand` – the manufacturer or fashion label (e.g., NIKE, ADIDAS, PUMA);
 - `name` – the catalog name or model title of the sneaker;
 - `description` – a longer free-text description of the product;
 - `color` – the textual color representation (e.g., “White/Black”);
 - `size` – the product size, typically in EU or UK format.
- **Visual and source information:** Links and metadata related to the product’s imagery and origin:

- `mainImage.original_url` – the URL for the primary product image;
- `secondaryImage.original_url` – optional URLs for additional images;
- Merchant and source identifiers describing the data origin within the marketplace.

Exploration of the dataset revealed that identical products frequently appeared multiple times across merchants, often with **inconsistent or incomplete meta-data**. For instance, brand and color fields were sometimes missing, textual attributes contained non-standard capitalization or multilingual entries, and product names varied slightly between sellers. This variability reflected the decentralized nature of data entry in online marketplaces and underscored the need for extensive preprocessing to **ensure consistency** before clustering and variant detection.

In addition to attribute inconsistencies, the JSON structure itself was deeply nested, with product attributes, image links, and merchant data stored across several hierarchical levels. Prior to analysis, it was therefore necessary to explore these nested relationships to identify which attributes were most informative for product identity and to prepare the data in a structured format suitable for multimodal learning.

5.2 Dresses Dataset – Used for Feature Extraction

The dataset used for Task 3 consisted of women’s dress product listings sourced from the Jinius fashion catalog. Like the sneakers dataset, it was provided in a nested JSON format and contained a combination of descriptive, categorical, and administrative attributes alongside image links. Each record represented an individual product listing submitted by a merchant and served as the basis for visual feature extraction experiments.

The dataset included a wide range of attributes describing each dress, which can be grouped into the following categories:

- **Identification fields:** Unique administrative identifiers that enabled each product to be traced within the marketplace system:
 - `mirakl_product_id` – the global product identifier within the Jinius platform;
 - `product_sku` – the merchant-specific stock-keeping unit used internally by sellers.
- **Descriptive attributes:** Fields describing the product’s visual and categorical characteristics:
 - `brand` – the manufacturer or fashion label (often missing or inconsistently provided);

- **category** – the product category (in this case mostly "FaWoCldresses", indicating Fashion Women's Clothes Dresses);
 - **name** [en_GB] – the English product name, often containing model or style references;
 - **shortDescription** [en_GB] – a short textual summary of the dress, typically including material or design information;
 - **color** – the dominant color or color combination;
 - **size** – available size options (textual or numeric).
- **Image and source information:** Fields linking each product to its corresponding images and merchant source:
 - **mainImage.original_url** – the primary image link for the dress;
 - **secondaryImage.original_url** – optional secondary or alternate images, when available;
 - Merchant and catalog identifiers describing the product's origin within the marketplace.

Exploration of this dataset revealed **similar inconsistencies** to those observed in the sneakers data. Many listings lacked complete information, particularly in the **brand** and **color** fields, and textual descriptors varied significantly in format and language. Image-related attributes also required validation, as some links were missing, duplicated, or contained images with non-relevant backgrounds. These issues emphasized the importance of thorough preprocessing before applying computer vision models.

Additionally, the nested JSON format introduced structural complexity, with product attributes, merchant metadata, and images located at different hierarchical levels. Exploring these relationships was necessary to identify the most relevant attributes for visual representation and ensure data consistency. Through this exploration, it became clear which fields were most informative for defining visual and semantic product characteristics, guiding the preprocessing, feature extraction, and modeling steps that followed.

Overall, the dresses dataset provided a rich yet complex source of multimodal data. Its combination of textual and visual attributes made it well-suited for experiments involving vision-language models such as CLIP, but also highlighted the need for **data cleaning**, **normalization**, and careful **attribute selection** prior to analysis.

Chapter 6

Data Preprocessing and Exploratory Data Analysis (EDA)

6.1 Sneakers Dataset

As mentioned already, the original dataset was provided in JSON format, containing several nested fields. For instance, the field `data` in each record included a dictionary with subfields such as `size`, `brand`, and `color`. To facilitate inspection and subsequent processing, the dataset was restructured into three separate DataFrames, and all nested fields were flattened. During this transformation, certain attributes that were irrelevant to the project objectives were excluded (e.g., the creation date field, which is used internally by Jinius as an additional information to the product).

Specifically, three distinct DataFrames were constructed:

- A **metadata DataFrame**, containing product identifiers such as the Mirakl product ID (an internal identifier used within the Mirakl platform).
- A **data DataFrame**, containing key product characteristics such as brand and color.
- A **sources DataFrame**, holding information about the origin of each listing, including provider ID and related attributes.

The dataset comprised a total of 5,823 sneaker products. An exploratory data analysis (EDA) revealed a considerable presence of missing values across several attributes, including fields crucial to the subsequent analyses. Notably, 608 products lacked a product name in Greek, though all had English names. Similarly, 1,336 products were missing a short Greek description, while none were missing a short English description. In contrast, 5,585 long English descriptions and 5,716 long Greek descriptions were missing. Consequently, all subsequent analyses focused exclusively on the English name and short English description fields.

Further inspection showed that 11 products were missing brand information and 15 had no color specified. Regarding product images, three separate URL

fields were available. However, several entries contained missing or broken links. To ensure valid image inputs for embedding generation, the preprocessing pipeline sequentially verified the existence of each image URL: the first image was used if available; otherwise, the second or third was used in order of priority. If all three URLs were missing or invalid, the corresponding product was excluded from subsequent analysis.

Finally, an initial inspection identified 128 unique sneaker brands. Upon closer examination, several inconsistencies in spelling and capitalization were observed (e.g., “EA7” vs. “Emporio Armani”), leading to duplicate brand entries. After standardizing brand names, the correct number of unique brands was determined to be 121.

Finally, a visualization function was implemented and employed across all subsequent approaches to facilitate the inspection of clustering results. This function enabled the visual examination of product images within each cluster, providing valuable qualitative insights into the coherence and accuracy of the generated groupings.

6.2 Dresses Dataset

The dresses dataset was likewise provided in JSON format and contained several nested fields. The field structure was consistent with that of the sneakers dataset, as Jinius maintains a standardized schema for all fashion-related products. However, compared to the sneakers dataset, the dresses data included category-specific attributes that reflected differences in product characteristics. For instance, the sneakers dataset contained a heel height field, whereas the dresses dataset featured attributes such as sleeve length, which are more relevant to apparel.

Accordingly, the dataset was transformed in the same manner as before, being split into three separate DataFrames (metadata, data, and sources) and the nested fields were flattened.

The dataset comprised a total of 32,512 dress products. The exploratory data analysis revealed missing values across multiple attributes. Specifically, 20 products lacked brand information, 1,760 had missing color, and 27,355 had missing Greek names, while none were missing English names. Similarly, 25,392 products lacked a short Greek description, though all had an English short description. Fourteen products were missing their main image, and the majority of products (over 32,000 entries) were missing long descriptions in both English and Greek.

Chapter 7

Methodology

7.1 Approaches for Identical/Variant Detection (Tasks 1 and 2)

7.1.1 Approach 1: Metadata-based Grouping

Methodology

The first approach attempted to detect identical products by applying a `groupby` operation over a strict set of metadata fields. The underlying assumption was that two products should be considered identical if and only if they matched exactly in their brand, color, category, name, and description attributes. By grouping on this composite key, any products with identical values across all five fields were placed in the same cluster. This approach relied entirely on the completeness and consistency of the available metadata, assuming that identical items would be described in precisely the same way across all listings. [1]

Results and Findings

Although straightforward to implement, this approach proved to be highly ineffective in practice and exposed the limitations of relying solely on raw metadata. The method was overly rigid since products that were clearly the same model but differed slightly in a single attribute, such as color, were placed in separate clusters. This rigidity resulted in the systematic exclusion of true variants. Metadata inconsistency also posed a major challenge because even small differences in product names or descriptions, such as variations in capitalization, spacing, or merchant specific wording, caused identical products to be separated into different groups. Furthermore, many records contained incomplete attribute information, which fragmented the clusters and reduced overall reliability. As the method lacked tolerance for noise or approximate similarity, every mismatch was treated as decisive, regardless of how minor the difference might have been. In practical terms, the approach achieved perfect precision only in cases where the metadata matched exactly, but its recall was extremely poor. Very few duplicate or variant products were successfully grouped, and the resulting clusters lacked robustness

and generalization. [1] Consequently, this approach served primarily as a baseline to demonstrate the limitations of exact equality checks and to motivate the adoption of more flexible embedding based methods in subsequent experiments.

7.1.2 Approach 2: Embedding-based Clustering

Methodology

In this approach, product images were processed to generate visual embeddings, while structured attributes (brand, color, shoe type) and product name were transformed into textual embeddings. Visual embeddings were extracted using ResNet50 [3], whereas attribute embeddings were obtained through TF-IDF vectorization. [16, 4] The resulting representations were then used to compute pairwise similarities among products, which were subsequently grouped using the DBSCAN clustering algorithm. For each of these, multiple hyperparameter settings were explored in order to identify those that produced the most coherent clusters. In particular, different weighting schemes were applied when combining the similarity matrices derived from the three types of embeddings. Additionally, several values of the DBSCAN parameter **eps**¹ were tested in order to optimize the balance between cluster compactness and coverage. Several alternative strategies were also investigated. These included performing clustering based solely on image embeddings, employing Principal Component Analysis (PCA) to reduce the dimensionality of the feature space in order to mitigate noise prior to clustering, and applying HDBSCAN [15] as an alternative to DBSCAN for clustering, effectively eliminating the hard to tune hyperparameter **eps** and potentially providing qualitative improvement over DBSCAN. [22] Cluster quality was assessed through a combination of the Silhouette score, manual visual inspection, and evaluation against a manually labeled subset of the data. The evaluation was conducted on a manually annotated sample dataset, where labels corresponded to variant groupings. Consequently, only Task 2 (variant detection) could be directly assessed. In principle, Task 1 (duplicate detection) could be evaluated in a similar manner, provided a labeled dataset tailored to duplicate products were available. To account for **class imbalance** in the labeled data, **micro-averaged precision** was adopted as the **primary evaluation metric**. Unlike macro-averaging, which assigns equal weight to all classes, micro-averaging **treats each instance equally**, thereby mitigating distortions caused by imbalanced class distributions. This choice emphasizes the minimization of false positives, i.e., incorrectly grouped products, which is particularly critical for maintaining catalog integrity in marketplace applications.

¹**eps** defines the maximum distance between two samples for one to be considered as part of the neighborhood of the other.

Results and Findings

This approach faced several notable challenges. First, the image embeddings proved unreliable in some cases, as many product images could not be processed due to broken links or invalid content, thereby reducing the coverage of the method. Second, DBSCAN exhibited high sensitivity to the choice of the `eps` parameter, which significantly influenced clustering outcomes. Third, the clustering quality was mixed: in certain instances, unrelated products were erroneously merged into the same cluster (false positives), while genuine variants were incorrectly separated (false negatives). Additionally, a substantial number of products remained unassigned to any cluster, resulting in a high proportion of noise points. The best-performing configuration on the test set, achieved a micro-averaged precision of 0.026121 and a micro-averaged accuracy of 0.938794. This solution produced 108 clusters, while leaving 196 sneakers unclustered. The corresponding setup was as follows:

- Clustering algorithm: DBSCAN with `eps` = 0.36, `min_samples` = 2, and `metric` = precomputed.
- Similarity weights: 0.4 for product name embeddings, 0.5 for image embeddings, and 0.1 for structured attribute embeddings.
- Distance metric: 1 - similarity matrix.

The shortcomings of this approach motivated the development of more robust hybrid methods in later stages.

Approach 3: Hybrid Image–Text Embedding Clustering

Methodology

In this approach, only the product descriptions were vectorized using TF-IDF, which were then combined with the previously extracted image embeddings. Two integration strategies were explored: (i) **concatenation** and (ii) **element-wise multiplication**, allowing us to examine how different fusion methods influenced clustering performance. For concatenation, the combined embeddings were clustered using DBSCAN, K-means, and agglomerative clustering. An attempt to tune the hyperparameter K^2 for K-means was performed, using the elbow method based on inertia values. The tested values of K fell in the range of 1 to 100, but inertia values continued to decrease without showing a clear inflection point. Extending the range to $K = 800 - 900$ yielded the same behavior, as inertia still did not level off (see Figure 12.6). This observation, combined with the high computational cost of exploring larger values, prevented further investigation of K-means in this setting. For element-wise multiplication, clustering was again performed using DBSCAN, K-means, and agglomerative clustering. Similar to the concatenation experiments, tuning k for K-means failed to yield a clear elbow point, which limited the potential of this clustering method. As with the previous approach,

²The number of clusters to produce.

the quality of the clusters was assessed using Silhouette scores, manual visual inspection and evaluation on a labeled test set. Additionally, the final evaluation of the methods in this approach was performed only for Task 2 and was based on the labeled sample dataset.

Results and Findings

This approach encountered difficulties similar to those observed in the previous method, including challenges in hyperparameter tuning and the presence of invalid product images. The best-performing configuration on the test set achieved a micro-averaged precision of 0.0040 and a micro-averaged accuracy of 0.7971. This result was obtained using agglomerative clustering with the following settings:

- `n_clusters`³ = 10,
- `linkage`⁴ = 'ward',
- Embedding fusion strategy: element-wise multiplication of description and image embeddings.

Similarly, this approach proved highly insufficient for capturing product groupings, as it frequently failed to cluster products that belonged together while at times incorrectly merging unrelated items. These shortcomings underscored the necessity of developing a more effective method capable of achieving acceptable performance in terms of micro-averaged precision.

7.1.3 Approach 4: Combining Group-by Operations with Contextual Embeddings Hybrid Clustering

Methodology

The fourth approach investigated the use of **contextual metadata embeddings** in combination with image embeddings for clustering products into groups of identical listings and variants. This multimodal strategy employed Sentence-BERT for textual attributes [5] and a neural network pre-trained on ImageNet for visual features [3]. Clustering was performed using the HDBSCAN algorithm [15], which is capable of identifying clusters of varying density while treating noise points appropriately. Four sub-methods were designed, each introducing refinements in grouping logic or embedding transformations to evaluate the interaction between textual and visual features in defining product similarity. [16, 12, 13]

³The number of clusters to identify.

⁴The Ward linkage criterion minimizes the variance within the clusters being merged.

7.1.4 Method *

The first sub-method introduced a preliminary grouping step based on brand and category. [5] Within each group, textual embeddings were generated using Sentence-BERT, which captures semantic meaning across different product names and descriptions. Image embeddings were extracted using ResNet50 and combined with the textual embeddings through vector concatenation to form multimodal product representations. HDBSCAN [15] was then applied with a `min_cluster_size` of 2^5 and the Euclidean metric⁶. This approach constrained clustering to semantically coherent subsets, improving internal cohesion within each brand-category group.

7.1.5 Method **

The second sub-method increased the strictness of grouping by incorporating the product name (English field) alongside brand and category. This additional key reduced heterogeneity within groups, ensuring that clustering was performed only among closely related listings. Sentence-BERT and image embeddings were again concatenated to form a unified representation, and clustering was carried out using HDBSCAN with the same hyperparameters (`min_cluster_size` = 2, metric = Euclidean). The refinement of grouping keys reduced noise and led to more precise and internally consistent clusters that effectively captured product variants.

7.1.6 Method ***

The third sub-method aimed to identify identical products rather than variants. In this case, color was added to the grouping keys in addition to brand, category, and name. The rationale was that products sharing all these attributes were highly likely to represent the same physical item rather than variations. Sentence-BERT and image embeddings were concatenated within each group, and HDBSCAN was applied using `min_cluster_size` = 2 and the Euclidean metric. This stricter criterion aligned well with deduplication objectives, although the method could not be quantitatively evaluated due to the absence of labeled ground truth for identical products.

7.1.7 Method ****

The fourth sub-method examined the influence of color information on clustering behavior. It replicated the setup of Method ** but converted all product images to grayscale prior to embedding extraction. This preprocessing step removed color as a discriminative feature, forcing the algorithm to rely more heavily

⁵The parameter `min_cluster_size` defines the smallest grouping that HDBSCAN [15] considers a valid cluster. A value of 2 allows the detection of very small and fine-grained clusters, suitable for identifying pairs of identical or highly similar products.

⁶The Euclidean metric measures the straight-line distance between points in the joint embedding space, enabling the algorithm to identify compact and well-defined clusters of products.

on structural and textural cues when measuring similarity. Sentence-BERT embeddings were again combined with grayscale image embeddings and clustered using HDBSCAN [15] with `min_cluster_size = 2` and the Euclidean metric. The resulting clusters remained coherent and stable even without color information, although the performance did not exceed that of Method **, indicating that color contributes positively to distinguishing between product variants. Taken together, these four sub-methods constitute a systematic exploration of multi-modal clustering strategies. By varying grouping criteria and the role of color in visual embeddings, the methodology evaluated how textual semantics and visual similarity jointly influence the ability to identify duplicates and variants within marketplace data. [12]

Results and Findings - Cluster Visualisations

At this stage of the project, no ground truth labels were available for quantitative evaluation. To enable qualitative assessment, HTML-based visualization functions were implemented to render the contents of each cluster. This approach allowed for manual inspection of grouped products, facilitating comparison of visual and textual similarity and verification of whether clusters corresponded to meaningful product variants or duplicates. Visual inspection showed that Methods **, *** and **** yielded the most coherent and reliable clusters. However, formal evaluation using a labeled dataset was later necessary to confirm these findings quantitatively (see Figures 12.1, 12.2, 12.3, 12.4, 12.5). This approach ultimately produced the most robust and interpretable results and met the technical requirements for production deployment, representing a central outcome of the project.

7.2 Method Evaluation

7.2.1 Manual Labeling of Evaluation Sample

A manually labeled subset of the dataset was required in order to provide a reliable ground truth against which clustering algorithms could be evaluated. Since the original marketplace data contained inconsistencies, duplicates, and variants without explicit annotations, an external validation set was essential for quantitative assessment. The purpose of constructing this sample was twofold: first, to establish reference clusters of product variants that could be compared to algorithmic outputs, and second, to enable the calculation of evaluation metrics such as precision and confusion matrices. Without this manually labeled subset, algorithmic results could only be examined qualitatively, which limited the ability to benchmark methods rigorously or to compare alternative approaches. By focusing the labeling effort specifically on product variant detection (Task 2), the evaluation process remained aligned with the project’s primary objective of grouping product variants consistently across multiple merchants. The creation of this ground truth sample followed a systematic **three-step process**. Initially, a random subset of

1,000 sneaker listings was selected to ensure coverage across brands, categories, and sellers. In the **first stage**, products were grouped based on visual similarity using CLIP image embeddings. Images were preprocessed, embedded, and compared pairwise with cosine similarity, and products with highly similar embeddings (above a strict threshold) were merged into connected components representing potential variant groups. In the **second stage**, products were grouped by product name in English, thereby capturing identical or near-identical naming conventions across merchants. Finally, in the **third stage**, textual descriptions were vectorized using TF-IDF and compared at high similarity thresholds to capture cases where naming differed but descriptive content indicated the same product family. Across these steps, **the resulting clusters were manually inspected to confirm their validity**, ensuring that mislabeled or noisy examples were corrected. This layered design, combining automated grouping with manual oversight, produced a robust labeled sample that could subsequently be used for quantitative evaluation of clustering performance.

7.2.2 Evaluation Metrics

Evaluation of clustering quality was based on three complementary metrics: **micro-average within-cluster precision**, **micro-average accuracy**, and **cluster purity**. Each of these metrics captures a different aspect of clustering performance, and together they provided a comprehensive assessment of how effectively products were grouped into variants. Micro-average within-cluster precision was employed to quantify the correctness of positive associations made by the clustering algorithm. This metric aggregates pairwise predictions across all clusters and weights them according to class frequency, thereby addressing the inherent imbalance present in marketplace data, where some product models are highly represented while others appear rarely. Precision is particularly relevant in this setting because minimizing false positives, that is, avoiding the incorrect merging of unrelated products, is essential for maintaining user trust and catalog integrity. Micro-average accuracy was also calculated to provide an overall measure of correctness across all pairwise comparisons between predicted clusters and ground truth labels. This metric evaluates both true positive and true negative associations, thereby reflecting the extent to which the clustering reproduced the labeled relationships at the dataset level. Given the practical objective of accurately separating distinct product families while grouping their variants, micro-average accuracy offered a reliable summary measure of end-to-end performance. **Cluster purity** was included as an intuitive measure of how homogeneous the clusters were with respect to the labeled ground truth. Purity is computed by assigning each predicted cluster to the most common ground truth class within it and calculating the proportion of correctly assigned items across all clusters. In the context of product variant detection, a high purity score indicates that clusters are internally consistent and dominated by a single product family, which is critical for ensuring a seamless marketplace browsing experience. These three metrics were computed directly from the manually labeled evaluation subset. Pairwise associations between products were derived from the ground truth labels and compared against

the predicted clusters. Micro-average precision and accuracy were then obtained by aggregating these comparisons, while cluster purity was calculated by examining the distribution of ground truth labels within each predicted cluster. This combination of metrics ensured that the evaluation aligned with the project’s goal of identifying product variants reliably and minimizing incorrect merges across unrelated products.

7.3 Model Performance

As mentioned earlier, evaluation was only possible for the task of product variant detection, since the manually labeled ground truth dataset was specifically created for that purpose. Methods developed for identical product detection could not be quantitatively assessed due to the absence of corresponding labeled data. Therefore, only the approaches designed for variant clustering were evaluated using the available ground truth. Despite extensive experimentation with different combinations of metadata fields, textual embeddings, image embeddings, and clustering algorithms, none of these earlier methods achieved satisfactory clustering performance. Quantitative scores remained low, and the resulting clusters showed poor alignment with the manually grouped products. These weaknesses were further confirmed through manual visual inspection, which revealed that many clusters either merged unrelated products or failed to group clear variants together. This reinforced the conclusion that purely metadata-based or non-contextual embedding approaches were insufficient for the complexity of real marketplace data. However, the methods developed under Approach 4 yielded promising results, with evaluation metrics approaching near-perfect scores across all measures.

Chapter 8

Feature Extraction from Dress Images (Task 3)

8.1 Methodology

The third task focused on **identifying descriptive fashion attributes**, such as sleeve length, neckline type, pattern and dress occasion, directly from women’s dress images in the Jinius dataset. Unlike Tasks 1 and 2, which relied on meta-data and clustering, this stage explored how visual-language models could infer attributes without supervised training.

The approach was based on the CLIP model developed by Radford et al. [6], which aligns image and text representations within a shared embedding space by jointly training on large-scale image-text pairs. This enables classification through image-text similarity rather than explicit labels. Therefore, CLIP was selected for this task because it can perform zero-shot recognition, hence identifying attributes it has never been explicitly trained on, making it ideal for datasets like Jinius, where detailed annotations are unavailable. Clothing attribute prediction is identified as a core mid-level fashion understanding task [2], emphasizing the importance of attributes such as sleeve type, pattern, material, and shape; precisely the features targeted in Task 3. Building on the findings of Baron et al. [18], who showed that concise attribute-focused descriptions improve fine-grained recognition, CLIP was applied here to examine how effectively it could extract dress attributes such as sleeve type, neckline, pattern and occasion directly from product images.

Each image was encoded using CLIP’s visual encoder, while textual descriptions representing potential dress attributes were transformed into embeddings using the corresponding language encoder. Cosine similarity was then computed between the image and each textual prompt, and the description with the highest similarity was returned as the predicted label. This procedure allowed attribute classification entirely in a zero-shot manner, without any model fine-tuning.

8.1.1 A/B Testing of Prompt Design

Because CLIP’s performance depends heavily on how attributes are phrased, **two prompt formulations were evaluated** to determine which more accurately represented visual concepts. The first set consisted of short, attribute-focused phrases (e.g., “a sleeveless dress” or “a floral dress”), while the second contained longer, narrative sentences (e.g., “This dress is sleeveless, showing the shoulders.”). All images were tested under both configurations.

The shorter prompts consistently produced more stable predictions, particularly for attributes with clear visual cues such as sleeve style or pattern. This observation is consistent with previous work showing that brief phrasing improves zero-shot image–text alignment [18, 19, 20]. Longer prompts tended to introduce irrelevant linguistic detail, weakening the alignment signal between text and image embeddings.

8.1.2 Garment Localization with Object Detection

To reduce the influence of irrelevant background elements and improve focus on the clothing region, an object-detection stage was added before CLIP inference. Gu et al. [2] highlight that occlusion, background clutter, and pose variation significantly hinder fashion attribute recognition, reinforcing the importance of garment-level localization before analysis. Following the approach of Chang et al. [21], the YOLO algorithm [7] was used to detect the person or dress area within each image. The detected bounding box was used to crop the image tightly around the garment before passing it to CLIP.

This preprocessing step minimized background interference and improved recognition consistency across diverse lighting and pose conditions.

8.1.3 Results and Findings

The final pipeline, being the combination of YOLO-based cropping with CLIP inference using short, attribute-focused prompts, yielded reliable qualitative results. The system correctly identified prominent features such as sleeve length, overall style, and common patterns across most test images (see Figure 12.7). Nevertheless, several limitations were observed. Fine-grained elements such as neckline classification remained challenging, especially when visual cues were partially obscured by accessories or hair. Gu et al. [2] similarly emphasize that fine-grained clothing attributes are difficult to recognize due to occlusion, non-rigid garment deformation, and the small spatial regions in which these attributes appear. In addition, the restricted range of prompt categories constrained the diversity of attributes that could be detected.

Overall, the results validate earlier research on the importance of prompt design and region localization in zero-shot vision–language modeling. Integrating concise prompt engineering with detection-based cropping substantially improved robustness and interpretability compared to direct image input. Future work could extend this approach by expanding the prompt library, filtering accessory inter-

ference, or experimenting with domain-specific models such as CALIP [19] and FashionCLIP [20], which are optimized for fine-grained attribute recognition in fashion contexts.

Chapter 9

Results and Discussion

9.1 Tasks 1 and 2

The results for the product and variant detection tasks revealed a clear progression in performance across the four proposed approaches. Each successive method incorporated additional levels of semantic and visual understanding, moving from simple metadata grouping to multimodal clustering with contextual embeddings.

9.1.1 Results

Approach 1, which relied exclusively on exact metadata matching, provided a reliable but overly rigid baseline. It achieved perfect precision for listings with identical attribute values but failed to generalize to cases with incomplete or inconsistent metadata. As a result, the recall rate was extremely low, and only a small fraction of duplicates or variants were successfully grouped. This confirmed that strict equality-based methods are inadequate for large, heterogeneous marketplaces where merchant input is not standardized.

Approach 2 introduced embedding-based clustering using ResNet50 image embeddings and TF-IDF representations of textual attributes. Although this method demonstrated conceptual improvement by capturing semantic similarity, it suffered from practical limitations. Image coverage was reduced due to broken links, and DBSCAN exhibited high sensitivity to hyperparameters, leading to unstable clustering outcomes. The best configuration reached a micro-averaged precision of only 0.026 and an accuracy of 0.939, confirming that unsupervised embedding fusion alone was insufficient to produce meaningful variant clusters.

Approach 3 attempted to integrate textual and visual signals more tightly through hybrid fusion strategies (concatenation and element-wise multiplication) followed by clustering with DBSCAN, K-Means, or agglomerative methods. However, both fusion strategies produced noisy and inconsistent groupings. The best-performing setup, based on agglomerative clustering, yielded a micro-averaged precision of 0.004 and an accuracy of 0.797, indicating that simple fusion of heterogeneous embeddings without contextual alignment cannot capture the nuanced similarities required for marketplace-level product grouping. Visual inspection

corroborated these findings, as many clusters merged unrelated items or failed to connect visually obvious variants.

Approach 4 produced a decisive performance improvement by integrating Sentence-BERT textual embeddings with ResNet50 visual embeddings, clustered using HDBSCAN [15]. The introduction of contextual representations allowed the model to interpret the semantics of product names and descriptions rather than relying solely on surface-level word similarity. Across its four sub-methods, clustering quality and interpretability improved substantially.

Qualitative inspection showed that Methods ** and **** yielded the most coherent clusters, effectively grouping variants differing only in attributes such as color or size. Method *** provided a stricter definition of identity by including color as a grouping key but was not quantitatively evaluated due to the lack of ground truth.

Quantitative evaluation on the manually labeled ground-truth sample (Table ??) confirmed these qualitative trends. Method ** achieved near-perfect results with a cluster purity of 0.9965, micro-average precision of 0.9700, and accuracy of 0.99995. Method **** also performed strongly, with purity and precision above 0.90. These results demonstrate that contextual multimodal embeddings significantly improve robustness and alignment with human judgment compared to earlier approaches.

9.1.2 Discussion

The progression in results across the four approaches reflects the broader trends observed in recent research on product matching and variant clustering. Similar to the findings of Peeters et al. [13], this study confirmed that contextual language models such as Sentence-BERT substantially improve product similarity estimation [16, 12, 13] compared to traditional TF-IDF or keyword-based methods. While Peeters et al. [13] focused on text-only representations, the present work extends this principle to a multimodal setting, integrating textual and visual embeddings to achieve greater robustness in noisy marketplace data.

The improvement achieved by Approach 4 also aligns with the conclusions of Li et al. [16], who demonstrated that combining visual and textual encoders in a dual-embedding framework enhances the discrimination of visually similar but semantically distinct products. However, whereas Li et al. concentrated on cross-platform matching, this work applies the concept within a single marketplace, addressing intra-catalog duplication and variant grouping. The near-perfect evaluation scores achieved by Method 4** indicate that multimodal fusion is not only effective across different domains but also generalizes well to real-world e-commerce environments where data inconsistencies are common.

In contrast, the limited performance of early methods based on raw metadata or simple embeddings reinforces the conclusions drawn by Shah et al. [17], who emphasized that shallow feature representations struggle with the complexity of product data. Likewise, the unstable clustering results of Approaches 2 and 3 mirror the challenges identified by Peeters and Bizer [14], who noted that embeddings without explicit contextual alignment often fail to separate semantically

distinct products. The results of this project therefore provide empirical evidence supporting these observations and demonstrate the necessity of contextual and multimodal learning for scalable product grouping.

Overall, the results bridge the gap between existing academic approaches and practical marketplace applications. Whereas most prior studies evaluated their models in controlled or cross-platform settings, this project successfully operationalized a multimodal clustering framework directly on production-scale marketplace data. The findings thus confirm that modern embedding architectures, when combined with density-based clustering such as HDBSCAN [15], can deliver performance levels suitable for real-world deployment in commercial environments.

9.2 Task 3

9.2.1 Results

The combined CLIP + YOLO pipeline demonstrated strong qualitative performance in recognizing high-level fashion attributes such as sleeve style, dress pattern, and overall occasion type. When tested on the Jinius dress dataset, CLIP accurately identified visually distinct attributes like “floral dress”, “short-sleeve dress” and “evening dress”. The inclusion of YOLO-based cropping markedly improved visual focus by eliminating large background areas, reflections, and irrelevant objects. This localization step led to more consistent attribute predictions across images with varying lighting conditions and poses, confirming the benefit of preprocessing through region detection.

In addition, A/B testing of prompt sets revealed that short, attribute-specific phrasing consistently outperformed longer descriptive sentences. Predictions generated using brief prompts were not only more accurate but also more stable across repeated runs. These outcomes align with findings by Baron et al. [18], Guo et al. [19], and Patrick et al. [20], who showed that compact prompts yield stronger text–image alignment and better zero-shot transfer in fine-grained recognition tasks.

However, several challenges persisted. Fine-grained details such as neckline shape or sleeve length were often misclassified, particularly when occluded by hair, accessories, or overlapping objects. In addition, the restricted prompt library limited the model’s descriptive scope, occasionally forcing it to select the nearest available attribute even when none matched perfectly. Because the dataset lacked verified ground-truth labels for dress attributes, quantitative metrics such as accuracy or F1-score could not be computed, and evaluation remained qualitative.

9.2.2 Discussion

The results of this task reinforce broader research trends in zero-shot vision–language modeling for fashion. Consistent with the conclusions of Baron et al. [18], concise prompts enhanced the discriminative power of CLIP by direct-

ing attention toward core visual features instead of peripheral linguistic context. In contrast, long descriptive sentences diluted the embedding signal by including words unrelated to the target attribute, confirming that prompt engineering is a critical design factor for zero-shot inference.

The observed improvement from YOLO-based cropping supports prior studies by Chang et al. [21], which highlight the importance of region localization in fashion attribute prediction. By constraining CLIP’s attention to garment-specific regions, the pipeline effectively reduced background noise and improved the consistency of predicted features. This supports the growing view in fashion computer vision that combining detection and recognition gives better results than analyzing the whole image directly.

Nonetheless, the absence of annotated attribute labels in the Jinius dataset limited the ability to perform numeric evaluation or cross-validate results statistically. While the qualitative inspection provided useful insights, future analyses would benefit from a manually labeled validation set to quantify precision and recall. Moreover, CLIP’s general-purpose training on web imagery, rather than fashion-specific data, partly explains the difficulties in recognizing subtle garment details, being a limitation similarly reported by Guo et al. [19] and Patrick et al. [20].

In conclusion, this task in the study shows that integrating prompt engineering and region-based detection substantially enhances the interpretability and stability of zero-shot fashion attribute recognition. The findings extend the current literature by applying these methods to a real-world e-commerce dataset rather than curated research collections, validating their practical relevance. Future research should therefore focus on constructing labeled attribute datasets for quantitative benchmarking, expanding the prompt vocabulary, and fine-tuning vision–language models such as FashionCLIP or CALIP on domain-specific fashion imagery to capture fine-grained attributes with higher precision.

Chapter 10

Conclusion

This project presented a complete framework for addressing the problem of duplicate and variant product listings in online marketplaces through multimodal clustering and feature extraction. By combining textual and visual information, the proposed methods improved the consistency, scalability, and usability of large product catalogs.

Tasks 1 and 2 addressed the identification and grouping of identical and variant products within the Jinius Marketplace catalog. The progression of approaches developed throughout the project demonstrated a clear evolution from simple attribute-based grouping to advanced multimodal clustering capable of understanding both textual and visual product relationships.

The early methods, relying solely on metadata and static embeddings, highlighted the limitations of purely rule-based or shallow representations in real-world marketplace data. They lacked robustness against incomplete or inconsistent attributes and failed to capture the semantic relationships between similar listings. In contrast, the final multimodal framework successfully overcame these challenges by leveraging contextual textual embeddings from Sentence-BERT in combination with visual embeddings from convolutional neural networks. The use of HDBSCAN further improved stability by identifying clusters of variable density without the need for predefining their number.

The results confirmed that contextual and multimodal representations substantially enhance the ability to distinguish between duplicates and variants, achieving near-perfect cluster purity and precision. These findings align with trends in the literature, reinforcing the effectiveness of contextualized embeddings for semantic understanding and demonstrating the added value of incorporating image information for variant-level differentiation.

Beyond achieving strong quantitative performance, the implemented approach provided meaningful qualitative improvements. Manual inspection showed that clusters corresponded closely to human judgment, effectively grouping products that shared the same model while distinguishing those differing in key attributes such as color or size. The resulting system not only met the company’s operational requirements but also offered a scalable, production-ready solution for ongoing product management.

Tasks 1 and 2 established a robust foundation for automated catalog orga-

nization in online marketplaces. By validating the feasibility and accuracy of multimodal clustering on real marketplace data, this work bridges the gap between academic research and industrial deployment, supporting efficient, scalable, and intelligent catalog management within the Jinius ecosystem.

The practical outcome of this work extends beyond experimental validation. The final methodology, based on contextual and visual embeddings, was deployed into the Jinius Marketplace production environment to automate product grouping and enhance catalog coherence. This deployment demonstrates the feasibility of translating research-driven methods into functional tools for e-commerce platforms.

The findings from Task 3 highlight the potential of vision-language models such as CLIP for enhancing product understanding in e-commerce settings where annotated data are limited. By combining prompt engineering with YOLO-based garment localization, the system was able to extract key visual attributes, such as sleeve length, neckline, and pattern, directly from product images in a zero-shot manner. Hence, this approach demonstrated that meaningful attribute recognition can be achieved without supervised training, offering a scalable solution for enriching catalog metadata. While evaluation was qualitative due to the lack of labeled ground truth, the experiments confirmed that concise, attribute-focused prompts and localized image inputs substantially improve reliability.

Overall, this task extends the clustering and variant-detection work of Tasks 1 and 2 by introducing an interpretable, vision-driven component that strengthens product representation and supports automated categorization within the Jinius Marketplace. Together, the three tasks form a cohesive end-to-end framework that enables duplicate and variant grouping while enriching product descriptions through visual analysis. This integrated system improves catalog transparency, enhances search and recommendation functionality, and lays the groundwork for future developments in intelligent, data-driven e-commerce management.

Chapter 11

Limitations and Future Work

Several limitations were identified throughout the development and evaluation of this project. These methodological constraints reflect broader challenges in fashion AI noted by Gu et al. [2], including inconsistent attribute annotations, limited availability of ground-truth labels, and the difficulty of evaluating fine-grained clothing attributes at scale. Their review emphasizes that both dataset fragmentation and weak labeling significantly restrict the performance of current fashion recognition systems, mirroring the limitations observed in this project. One key limitation concerns the size and representativeness of the manually labeled evaluation sample. A relatively small subset of the dataset was used to construct the ground truth, which may not fully capture the diversity and complexity of the complete marketplace catalog. As a result, the evaluation results, while indicative of model performance, may not generalize perfectly across all product categories and merchant listings. Additionally, Task 1, which focused on identifying identical products, could not be quantitatively evaluated due to the absence of a corresponding labeled dataset.

Another limitation involves the embedding representations used during the development and evaluation phases. When the methodologies for Tasks 1 and 2 were initially developed, ResNet50 image embeddings were employed, supplemented by a fallback mechanism for handling missing or invalid images. However, during the ground truth generation process, CLIP embeddings were identified as providing stricter and more discriminative visual representations, leading to improved clustering performance. Consequently, CLIP-based embeddings were loaded prior to evaluating the developed methods. While this change does not constitute a logical error, since both embedding types serve as valid visual representations, it does introduce a methodological inconsistency between the training and evaluation stages that should be addressed in future iterations.

In terms of model optimization, further improvements can be achieved through more extensive hyperparameter tuning of the clustering algorithms. Adjusting parameters such as the neighborhood radius in DBSCAN or the minimum cluster size in HDBSCAN could potentially refine the balance between precision and recall, yielding more stable and interpretable clusters. Future work may also extend Task 3, even though CLIP + YOLO pipeline provided promising qualitative re-

sults in extracting visual attributes from fashion imagery, several limitations were identified during experimentation.

A key limitation concerns the restricted scope of attributes and prompts available for evaluation. The predefined prompt library included only a small set of attribute types, mainly sleeve length, neckline, pattern, and occasion, which constrained the system’s descriptive capacity. The limited diversity of textual prompts also restricted the model’s ability to capture subtle variations between visually similar garments. This narrow vocabulary sometimes resulted in ambiguous predictions, as multiple attributes shared overlapping visual cues (for example, between short and three-quarter sleeves or between floral and abstract patterns).

Another significant limitation involved visual noise and occlusions in the dataset. Gu et al. [2] similarly highlight occlusion, pose variation, and background clutter as persistent obstacles in fine-grained fashion attribute prediction, underscoring that these issues are intrinsic to fashion imagery rather than specific to the dataset used in this study. While YOLO-based cropping reduced background interference, overlapping objects such as accessories, handbags, or hair frequently obscured parts of the garment, particularly the neckline region. These occlusions hindered the model’s ability to isolate relevant visual information, occasionally leading to incorrect attribute assignment. Variations in lighting, image angle, and resolution further affected the consistency of visual features extracted by CLIP’s encoder, which was trained on general image–text pairs rather than fashion-specific data. by enhancing the feature extraction pipeline through improved prompt engineering and the use of deeper or more specialized computer vision architectures for attribute recognition.

A further constraint was the absence of a labeled ground truth for attribute evaluation, which prevented quantitative performance measurement. The assessment was therefore limited to qualitative inspection rather than numeric metrics such as precision or recall. As a result, although the model’s predictions were visually convincing in many cases, their statistical reliability could not be objectively verified.

Future work should aim to expand and refine the prompt and attribute set, incorporating a broader range of garment characteristics such as fabric type, fit, and decorative details. Automating prompt generation through language models or adaptive template design could increase coverage and reduce human bias in wording. Additionally, fine-tuning CLIP or employing domain-specific vision–language models such as FashionCLIP [20] or CALIP [19] could enhance feature sensitivity to clothing-related attributes.

From a visual perspective, improved garment isolation using segmentation-based methods (e.g., Mask R-CNN or SAM) could minimize occlusion effects and ensure that only relevant clothing regions contribute to the embedding. Collecting a manually annotated subset of images with verified attribute labels would also enable quantitative evaluation and model benchmarking. Finally, integrating visual outputs with existing product metadata could provide a more comprehensive representation of each item, enhancing both marketplace searchability and catalog organization.

Concluding, the project's methodologies have already been transitioned into production as part of the Jinius Marketplace platform. Future work will therefore focus on seamless integration between the analytical pipeline and the existing web infrastructure. This involves collaboration with frontend and backend development teams to ensure seamless integration of the system within the live marketplace infrastructure, thereby enabling real-time product grouping and variant detection at scale.

Chapter 12

Appendix

Below are some examples of the outputs of HTML-based functions, showing the clusters generated from the methods in Approach 4.

Group 94 – Cluster 0 (191 items)

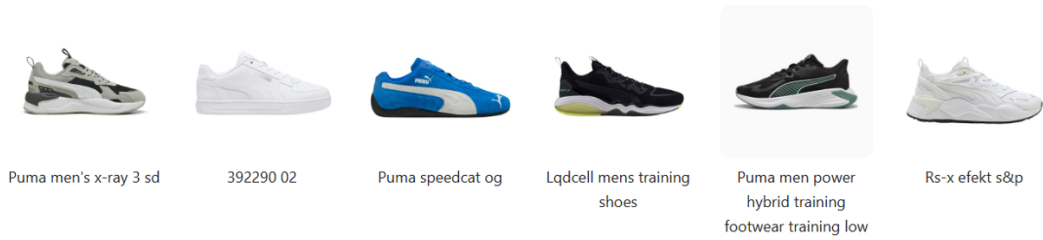


Figure 12.1: Example of a clustered product group produced by Method *. Visualized through the HTML-based inspection tool.

Group 539 – Cluster 1 (4 items)



Figure 12.2: Example of a clustered product group produced by Method ** using refined grouping keys.

Group 549 – Cluster 0 (9 items)

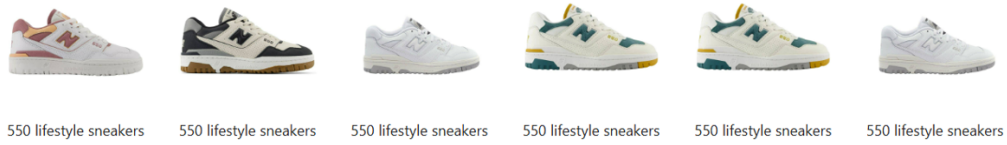
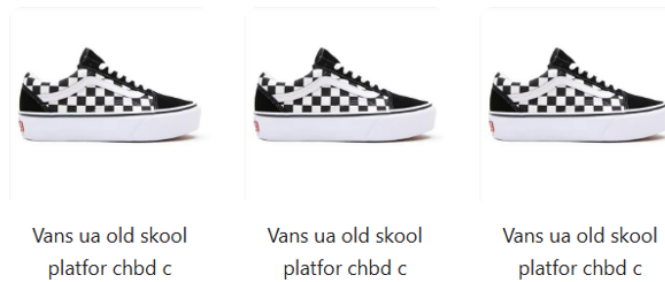


Figure 12.3: Additional example of cluster visualization for Method **, showing improved intra-cluster consistency.

Group 1 – Cluster -1 (3 items)



Group 2 – Cluster -1 (2 items)

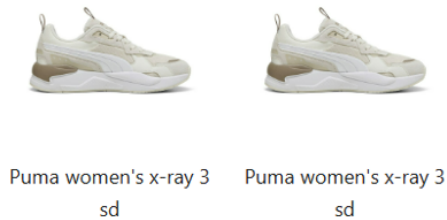


Figure 12.4: Cluster visualization for Method ***, corresponding to the identical product detection configuration.

group_105 – Cluster 1 (6 items)

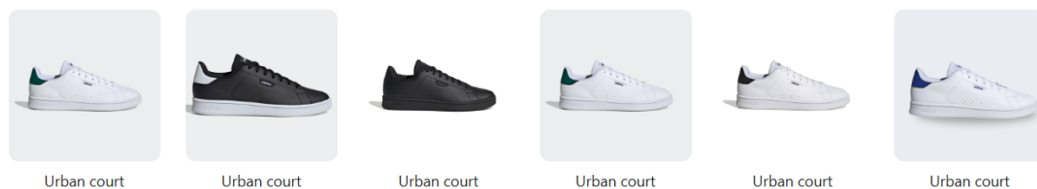


Figure 12.5: Cluster visualization for Method **** using grayscale image embeddings.

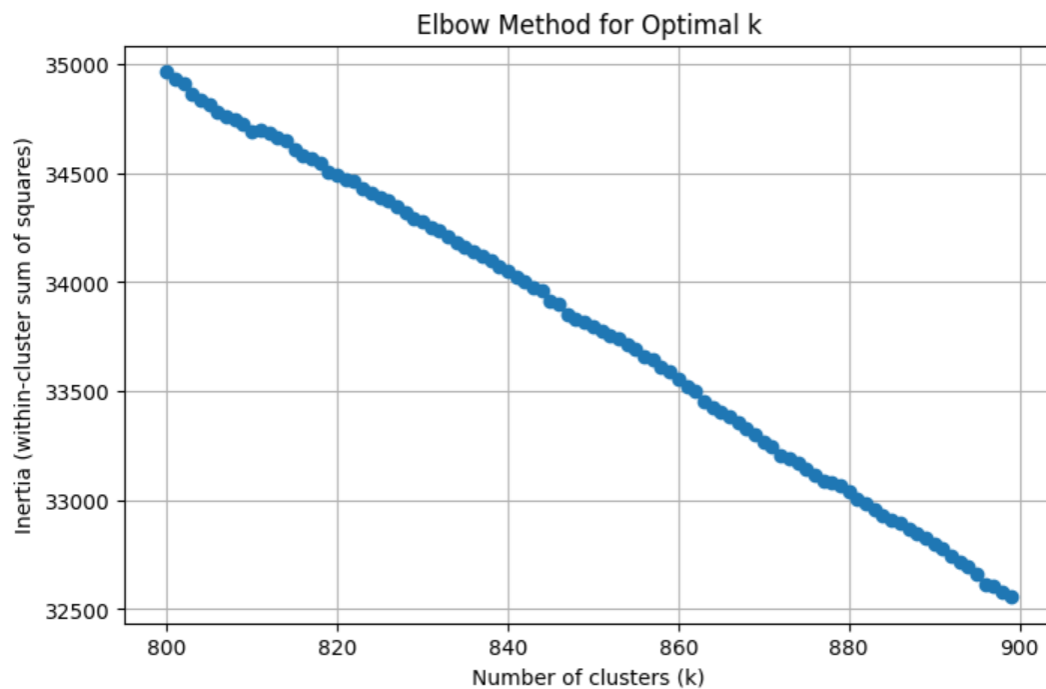


Figure 12.6: Elbow Method for K-means.



Figure 12.7: Example of YOLOv5 preprocessing applied to a dress image to remove background noise and improve CLIP-based visual attribute prediction.

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